02 Hashimoto-overlap PTC HLA susceptibility · HLA-II mediated dedifferentiation

Paper 2 HLA 는 *Hashimoto-overlap PTC* 전용이다. *Korean healthy/control baseline* 을 이용한 *6-allele exploratory OR/ Fisher/forest* 까지 만들었지만, 핵심 *caveat* 는 그대로다 – *Korean PTC* **는** *carrier frequency, baseline* **은** *published allele frequency* 이므로 이 통계는 *"metric-mismatched exploratory"* 로만 읽어야 한다.

`PTC values: 6/6 found`

`Statistics: exploratory only`

`Paper 4 GD: blocked from Paper 2`

`Author: Seungho Cook`

`Anchor: Paper 2 HT-isolated · Korean PTC n=874`

`Date: 2026-05-04`

I. 한 줄 결론

핵심: Paper 2 HLA = Hashimoto-overlap PTC 전용. 6-allele exploratory forest 실행됨. 단 PTC=carrier frequency, baseline=allele frequency 의 **metric mismatch** 를 숨기지 않는다.

PTC VALUES FOUND
6/6
Korean PTC Combined n=874 carrier frequencies

IN 2015 COVERAGE
5/6
A · B · C · DRB1 · DQB1 (no DPB1)

DPB1 PRIMARY PATH
1
AFND South Korea v2 + Jung 2023 cross-check

PAPER 4 GD REUSE
NO
Chu 2018 GD blocked from Paper 2

PROBLEM
HLA susceptibility 의 6-allele PTC vs Korean baseline 비교는 metric mismatch (carrier vs allele frequency) 를 가진다.

IDEA
Mismatch 를 숨기지 않고 exploratory layer 로 명시. PTC carrier count vs baseline 2n allele count 의 2x2 table 로 OR/Fisher/forest 를 산출.

DEFENSE
Paper 4 GD evidence 는 disease boundary 로 분리. Chu 2018 reactivation 없음. carrier-vs-allele 변환 없음.

DECISION
6-allele exploratory forest 완성. Yu 가 main/supplement/advisor-only placement 결정.

2. 비전공자 배경

이 페이지의 역할: Paper 1 페이지가 "lineage axis 를 왜 믿어도 되는지" 를 긴 설명으로 보여줬다면, 이 페이지는 "Hashimoto-overlap PTC 에서 HLA signal 을 어디까지 믿을 수 있고 어디서 멈춰야 하는지" 를 같은 방식으로 보여준다. 즉 결과만 던지는 페이지가 아니라, claim boundary 를 독자가 따라오게 만드는 설명형 pack 이다.

HLA란 무엇인가

HLA 는 면역세포가 "어떤 단백질 조각을 보여줄지" 를 정하는 유전적 표지다. 특정 HLA allele 은 자가면역 질환에서 항원 제시와 면역 반응의 방향을 바꿀 수 있다.

왜 HASHIMOTO-OVERLAP PTC인가

Paper 2 의 관심은 Graves/GD 가 아니라 Hashimoto thyroiditis 와 겹치는 PTC 다. 여기서는 HLA-II, antigen presentation, lymphoid/TLS, dedifferentiation 이 같은 축에서 정리된다.

HT와 GD는 왜 다른가

HT/HD 는 파괴성 갑상선염과 관련된 맥락이고, GD/Graves 는 자극성 자가항체와 갑상선 기능항진 맥락이다. 공유 HLA 가 있어도 disease mechanism 과 paper boundary 는 다르다.

현재 통계 GATE

PTC 쪽은 "개인이 allele 을 갖는가" 에 가까운 carrier frequency, baseline 문헌은 "2n allele 중 비율" 인 allele frequency 다. 이 차이를 승인 없이 통계 비교로 넘기지 않는다.

2.1 Reader glossary

TERM	쉬운 설명	PAPER 2 에서의 역할
HLA allele	면역세포가 항원을 보여주는 방식에 영향을 주는 유전형 표지.	PTC 가 Hashimoto-like immune context 와 만나는 genetic background 후보.
Carrier frequency	개인 n명 중 해당 allele 을 가진 사람의 비율.	Korean PTC pool 의 metric. 개인 단위다.
Allele frequency	2n개의 chromosome/allele copy 중 해당 allele copy 의 비율.	Korean baseline 문헌 대부분의 metric. allele 단위다.
Fisher exact test	작은 count table 에서도 쓸 수 있는 2x2 exact test.	이번 exploratory 통계에서 PTC carrier count 와 baseline allele count 를 비교하는 계산층.
Forest plot	여러 allele 의 OR 와 CI 를 한 눈에 보여주는 그림.	효과 방향과 불확실성을 시각화한다. clean case-control proof 는 아니다.
Paper 4 boundary	GD/Graves HLA evidence 는 별도 paper backlog 로 유지.	Chu 2018 GD signal 을 Paper 2 comparator 로 되살리지 않기 위한 방화벽.

3. 전체 논리 흐름

Paper 2 HLA 의 핵심은 "GD 에서 알려진 HLA 를 가져와 PTC 에 붙이는 것" 이 아니다. 먼저 Korean PTC pool 에서 6 개 target allele carrier frequency 가 있는지 확인하고, 그 다음 한국인 healthy/control baseline 에서 같은 allele 의 published allele frequency 가 있는지 확인한다. 마지막으로 metric mismatch 를 숨기지 않은 exploratory 통계로만 읽는다.

1. **Paper boundary 를 먼저 고정한다.** Paper 2 는 Hashimoto-overlap PTC/HLA-II dedifferentiation 이고, Paper 4 는 GD/Graves backlog 다.

2. **PTC carrier values 를 고정한다.** Korean PTC Combined n=874 에서 6 개 allele carrier count/frequency 를 읽는다.

3. **Korean baseline source 를 고른다.** In 2015 가 A/B/C/DRB1/DQB1 primary, AFND South Korea 가 DPB1 primary, Jung/Choe/Chung 은 cross-check 다.

4. **metric mismatch 를 공개한다.** carrier frequency 와 allele frequency 를 같은 것으로 바꾸지 않는다. 결과는 exploratory 로 라벨링한다.

5. **OR/Fisher/forest 를 실행한다.** PTC carrier count vs baseline 2n allele count 로 2x2 table 을 만들고 Fisher exact p 와 OR/CI 를 산출한다.

6. **claim 을 좁힌다.** "HLA susceptibility signal 후보" 까지 말하고, clean population-genetic case-control proof 나 GD-driven claim 은 말하지 않는다.

3.1 분석 사고 스토리

Q1 — 이게 PAPER 4 GD 분석인가?

아니다. GD rows 는 Paper 2 입력이 아니다. Paper 2 는 PTC carrier values 와 Korean healthy/control baseline 만 쓴다.

Q2 — 한국인 BASELINE 을 채울 수 있는가?

가능하다. In 2015 가 A/B/C/DRB1/DQB1 을 채우고, DPB1 은 AFND South Korea 와 Jung 2023 으로 방어한다.

Q3 — 통계를 바로 믿어도 되는가?

주의해야 한다. PTC 는 carrier, baseline 은 allele frequency 라 단위가 다르다. 그래서 결과는 exploratory layer 다.

Q4 — 그래도 왜 그리는가?

방향성과 규모를 advisor 가 빠르게 판단할 수 있다. Forest 는 의사결정 그림이지, 최종 manuscript claim 이 아니다.

Q5 — 무엇을 버렸는가?

Harbin Korean DRB1 fallback 은 sensitivity 로 내렸고, Chu 2018 GD 는 Paper 2 에서 재활성화하지 않았다.

FINAL — PAPER 2 HLA 는 실행 가능하지만 CAVEATED 다.

6 개 allele 모두 exploratory statistics 는 생성됐다. 다음 단계는 Yu 가 metric rule 과 wording boundary 를 승인하는 것이다.

3.2 Evidence ladder

LEVEL	EVIDENCE	READER SHOULD CONCLUDE	DO NOT CONCLUDE
1	Korean PTC carrier table n=874	PTC-side target allele values exist.	They are allele frequencies.
2	In 2015 Korean healthy baseline n=613	A/B/C/DRB1/DQB1 have direct Korean baseline values.	It covers DPB1.
3	AFND South Korea + Jung 2023 DPB1	DPB1*05:01 has a usable primary/cross-check path.	Jung full table has been fully archived.
4	Exploratory OR/ Fisher/forest	Direction/scale can be inspected.	Clean carrier-vs-carrier case-control inference is proven.
5	GD/Paper 4 evidence	Backlog registry only.	Paper 2 comparator or meta-analysis input.

4. Paper 2 vs Paper 4 boundary

TRACK	ALLOWED IN THIS PAGE	BLOCKED
Paper 2	Korean PTC, HT/AITD context, Korean healthy/control baseline, HLA-II mediated dedifferentiation	OR/Fisher/forest before Yu approval
Paper 4	Only mentioned as boundary/backlog	GD/Graves evidence as Paper 2 comparator, Chu 2018 reactivation
Paper 3	Frozen label only	ICI, HLA LOH, neoantigen

5. 6-allele source table summary

ALLELE	PTC METRIC	PTC VALUE	PRIMARY BASELINE	BASELINE METRIC/ VALUE	STATUS
A*02:07	carrier frequency	0.0812	In 2015	allele frequency 3.51%	needs Yu approval
B*46:01	carrier frequency	0.103	In 2015	allele frequency 5.06%	needs Yu approval
C*01:02	carrier frequency	0.2414	In 2015	allele frequency 17.81%	needs Yu approval
DPB1*05:01	carrier frequency	0.532	AFND South Korea current v2	allele frequency 0.3667	needs Yu approval
DQB1*02:01	carrier frequency	0.0	In 2015	allele frequency 2.12%	needs Yu approval
DRB1*07:01	carrier frequency	0.1144	In 2015	allele frequency 6.93%	needs Yu approval

Source TSV: `project/results/p2_pillar1_forest_v2/paper2_6allele_forest_source_table.tsv`

6. Exploratory OR/Fisher statistics

> **읽는 법:** 아래 통계는 실행됐다. 하지만 PTC row 는 carrier count 이고 baseline row 는 published allele frequency 에서 2n allele count 를 복원한 것이다. 따라서 "no conversion, metric-mismatched exploratory statistics" 로만 읽는다.

ALLELE	PTC CARRIERS/N	BASELINE ALLELES/2N	EXPLORATORY OR	95% CI	FISHER P	BASELINE SOURCE
A*02:07	71/874	43/1226	2.43	1.65-3.59	5.58e-06	In 2015
B*46:01	90/874	62/1226	2.16	1.54-3.02	7.38e-06	In 2015
C*01:02	211/874	218/1226	1.47	1.19-1.82	4.35e-04	In 2015
DPB1*05:01	465/874	499/1360	1.96	1.65-2.33	1.65e-14	AFND South Korea
DQB1*02:01	0/874	26/1226	0.026	0.0016-0.426	8.22e-07	In 2015
DRB1*07:01	100/874	85/1226	1.73	1.28-2.35	4.18e-04	In 2015

6.1 What this means

Exploratory statistics show that five alleles are higher in the Korean PTC carrier layer than the Korean baseline allele-frequency layer, while DQB1*02:01 is depleted in the PTC carrier layer. The strongest visual signal is DPB1*05:01 because it has both a large PTC carrier fraction and a Korean DPB1 baseline path.

6.2 What this does not mean

This does not prove a clean population-genetic case-control association, because the two rows of the 2x2 table are measured in different units. It also does not import GD/Graves biology into Paper 2, and it does not authorize a Paper 4 meta-analysis.

6.3 Files created

```
project/results/p2_pillar1_forest_v2/
paper2_6allele_or_fisher_primary.tsv
project/results/p2_pillar1_forest_v2/
paper2_6allele_exploratory_or_fisher_forest.png
project/reports/
2026_05_06_paper2_6allele_exploratory_stats_report.md
```

7. What changed in this package

PTC VALUES FOUND

6/6

All six target alleles have Korean PTC carrier-frequency values from the Combined rows.

IN 2015 COVERAGE

5/6

A, B, C, DRB1, DQB1. Does not cover DPB1.

DPB1 PRIMARY PATH

1

AFND South Korea v2 primary; Jung 2023 = Korean 11-locus cross-check.

EXPLORATORY TESTS

6

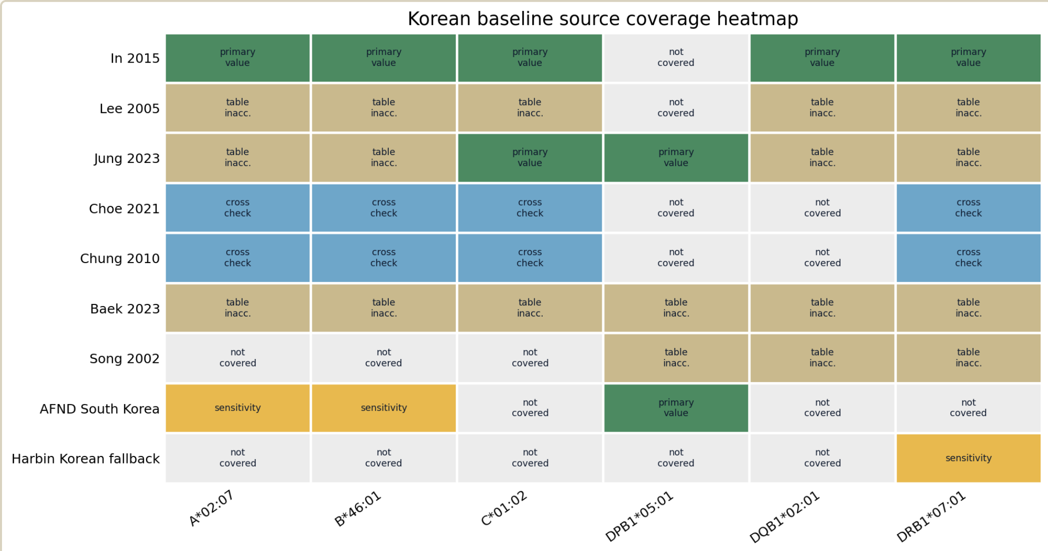
OR, Fisher p, forest executed for six alleles after approval. No Paper 4 meta-analysis.

8. Primary and cross-check sources

ROLE	SOURCES	MEANING
Primary candidate	In 2015 for A/B/C/DRB1/DQB1	Direct Korean healthy SBT allele-frequency table, n=613.
DPB1 primary	Current v2 AFND South Korea	Existing v2 DPB1 baseline remains primary for now.
DPB1 cross-check	Jung 2023	11-locus Korean NGS; abstract-level DPB1*05:01:01 = 35.1%, collapsed to 4-digit.
A/B/C/DRB1 cross-check	Choe 2021, Chung 2010	Useful for source consistency; not full 6-allele because DQB1/DPB1 are absent.
Sensitivity	Harbin Korean fallback for DRB1*07:01	Should not remain primary if direct Korean DRB1*07:01 values are used.

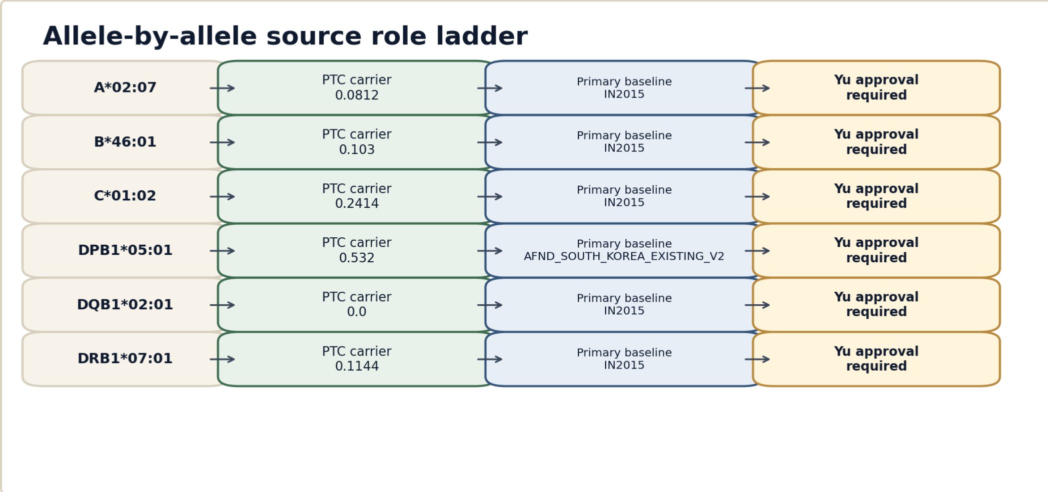
9. Figures (F1 ~ F15)

Click any figure to enlarge. F1–F12 define boundary/source/readiness. F13–F15 show the newly executed exploratory statistics with the metric caveat embedded.

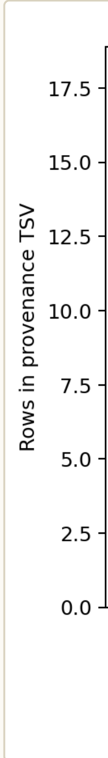


F3. What this means: In 2015 covers five alleles and Jung/AFND cover DPB1. What it does not mean: inaccessible tables were inferred.

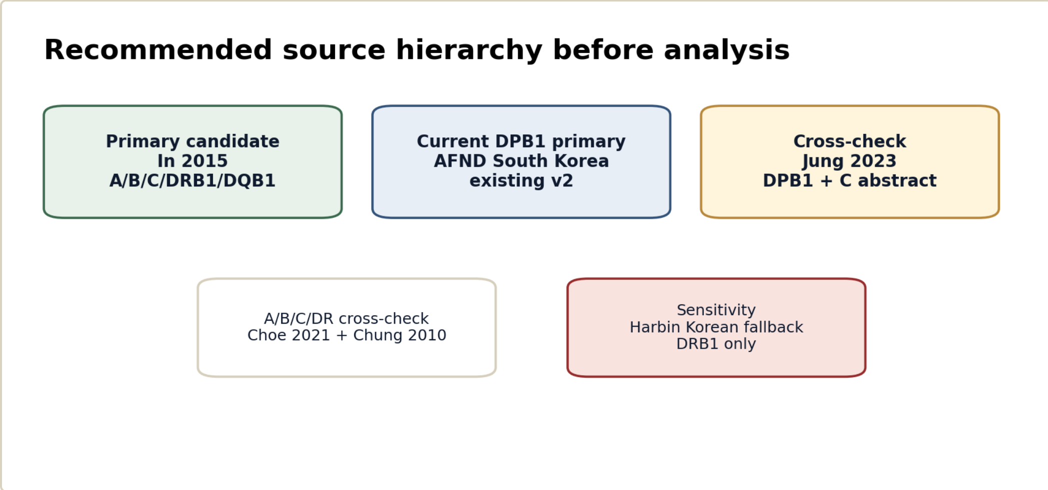
F4. Wha
measuren



F7. What this means: each allele now has a PTC value, primary baseline candidate, and approval gate. What it does not mean: the table has become an executable analysis.



F8. What
not mean



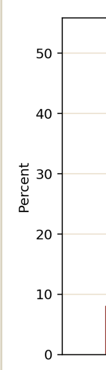
F11. What this means: source hierarchy is ready for advisor review. What it does not mean: hierarchy has Yu approval.

F12. Wh
hidden st

Paper 2 6-allele exploratory OR/Fisher table
PTC carrier count vs baseline allele count; no metric conversion

Allele	PTC carriers	Baseline alleles	OR	CI low	CI high	Fisher p
A*02:07	71/874	43/1226	2.43	1.65	3.59	5.6e-06
B*46:01	90/874	62/1226	2.16	1.54	3.02	7.4e-06
C*01:02	211/874	218/1226	1.47	1.19	1.82	0.00043
DPB1*05:01	465/874	499/1360	1.96	1.65	2.33	1.7e-14
DQB1*02:01	0/874	26/1226	0.0259	0.00158	0.426	8.2e-07
DRB1*07:01	100/874	85/1226	1.73	1.28	2.35	0.00042

F14. What this means: exact count basis and Fisher p are visible. What it does not mean: count reconstruction removes the metric caveat.



F15. Wh
does not

10. Expanded source / provenance tables

10.1 Six-allele readiness table

ALLELE	PTC SOURCE	PTC METRIC	BASELINE SOURCE	BASELINE METRIC	APPROVAL STATE
A*02:07	Korean PTC Combined n=874	carrier frequency	In 2015 Table 1	allele frequency	needs Yu approval
B*46:01	Korean PTC Combined n=874	carrier frequency	In 2015 Table 1	allele frequency	needs Yu approval
C*01:02	Korean PTC Combined n=874	carrier frequency	In 2015 Table 1	allele frequency	needs Yu approval
DPB1*05:01	Korean PTC Combined n=874	carrier frequency	AFND South Korea current v2	allele frequency	needs Yu approval
DQB1*02:01	Korean PTC Combined n=874	carrier frequency	In 2015 Table 1	allele frequency	needs Yu approval
DRB1*07:01	Korean PTC Combined n=874	carrier frequency	In 2015 Table 1	allele frequency	needs Yu approval

10.2 Source role table

SOURCE	ROLE	COVERAGE	CURRENT STATUS
In 2015	primary candidate	A/B/C/DRB1/DQB1	extracted
AFND South Korea current v2	DPB1 primary	DPB1*05:01	existing v2 source
Jung 2023	cross-check	DPB1*05:01 and C*01:02 visible in abstract	needs full table for broader use
Choe 2021	cross-check	A/B/C/DRB1	8-digit values collapsed with note
Chung 2010	cross-check	A/B/C/DRB1	indexed value extraction; archive full table before numeric reuse
Lee 2005 / Baek 2023 / Song 2002	defer or sensitivity	varies	table inaccessible in this pass

11. Claim boundary table

CLAIM	STATUS
Korean baseline source table prepared	YES
6-allele exploratory forest	EXECUTED — exploratory metric mismatch
C*01:02 fill	source available
DQB1*02:01 fill	source available
DPB1*05:01	existing AFND + Jung cross-check
Harbin fallback	sensitivity only
OR/Fisher/forest	EXECUTED after approval; caveated
Paper 4 GD	BLOCKED in Paper 2

12. Safe wording vs unsafe wording

TOPIC	SAFE WORDING	UNSAFE WORDING	REASON
HLA signal	Korean PTC shows exploratory enrichment/depletion patterns against Korean baseline sources.	HLA causally drives PTC.	The current layer is observational and metric-mismatched.
DPB1*05:01	DPB1*05:01 is the clearest candidate requiring clean metric confirmation.	DPB1*05:01 is definitively associated with PTC.	Baseline source and carrier-vs-allele distinction still matter.
DQB1*02:01	DQB1*02:01 is depleted in the PTC carrier table under this exploratory rule.	DQB1*02:01 is protective in Korean PTC.	Zero PTC carrier count needs careful typing/provenance review.
GD/Graves	GD HLA evidence belongs to Paper 4 backlog.	Chu 2018 GD validates Paper 2 HLA.	That would break the Paper 2/Paper 4 boundary.

13. Yu decision checklist · 한계 · 다음 행동

- [x] Q1 approve v2 scope and source table use for exploratory execution.
- [x] Q2 approve In/Jung/AFND baseline fill strategy for exploratory execution.
- [ ] Q3 decide whether 6-allele panel is primary or supplementary in final paper.
- [ ] Q4 decide whether metric-mismatched OR/Fisher can appear in main/supplement or advisor-only.
- [x] Q5 approve Harbin fallback as sensitivity only, not primary.

Next action: decide final placement: main figure, supplement, or advisor-only. Paper 4 remains frozen as registry unless separate approval is given.

14. 데이터 / 다운로드

📁 **P2_PILLAR1_FOREST_V2/**
6-allele forest source TSV + figures

📄 **STATS REPORT**
2026-05-06 exploratory stats report

→ **PAPER 4 GD HLA**
Pan-Asian Graves backlog (separated)

→ **PAPER 2 DETAIL A**
Hashimoto-overlap PTC core

→ **PAPER 2 DETAIL B**
Pillar continuations

← **HUB INDEX**
8-papers portfolio

Status. Paper 2 HLA = HT-overlap PTC only (per memory v18_paper2_HT_isolated). 6-allele exploratory forest executed after Yu Q1/Q2/Q5 approval. Paper 4 GD remains separate registry.

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paper2_hla.html`